Fiscal Year:	FY 2011	**Task Last Updated:**	FY 02/10/2014
PI Name:	Pawelczyk, Jim Ph.D.		
Project Title:	Improving Orthostatic Tolerance in Women: Control of Splanchnic and Cutaneous Vascular Capacitance		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
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Space Biology Special Category:	None		
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Zip Code:	16801-6900	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	05/15/2004	**End Date:**	09/30/2011
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	13
No. of Bachelor's Candidates:	13	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date is now 9/30/2011, per C. Guidry/JSC (from 9/30/2010)--(Ed. 10/26/2010) NOTE: End date is 9/30/2010, per C. Guidry/JSC (9/2010) NOTE: Received NCE to 9/30/2009 per J. Dardano/JSC (10/08) NOTE: Received NCE to 9/30/2008 per P. Denkins/JSC (10/07) NOTE: Received NCE to 9/30/2007 per PI (11/06)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Kenney, Larry (Pennsylvania State University) Leuenberger, Urs (Pennsylvania State College of Medicine) Williams, Nancy (Pennsylvania State University)		
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Task Description:	Clinical and empirical research reveals that orthostatic tolerance (OT) is less in women than men. This project seeks to relate sex differences in OT to biophysical properties of the circulation, and to validate physiological and pharmacological countermeasures to improve OT in women. In a gravitational field, cardiac filling is determined by the pressure difference between the right atrium and the hydrostatic indifferent point (HIP), the point in the circulation where pressure is unaffected by gravity. The HIP is the hydraulic equivalent of the center of mass of a solid object. We hypothesize that differences in anatomy and the distribution of vascular volume in women generate a HIP more caudal than men, causing greater falls in cardiac filling when standing. Accordingly, we will determine OT and the HIP in groups of women and men. Since the location of the HIP is dependent on regional vascular compliance, we will experimentally raise the HIP using a somatostatin analog or mechanical compression to reduce blood flow and vascular distension in the splanchnic circulation, while changing cutaneous compliance with passive skin cooling. These manipulations will produce measurable shifts in HIP, cardiac filling pressure, and OT. Moreover, they will establish the mechanism and means to ameliorate the greater incidence of post-spaceflight orthostatic intolerance and cardiovascular deconditioning in women by reducing vascular compliance in the two most distensible circulations in humans (splanchnic and cutaneous), shifting the HIP and vascular volume toward the heart. These new countermeasures can be adapted to spaceflight, and should prove applicable to populations of patients as well as orthostatically intolerant, but otherwise healthy, women.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research examined two different approaches to improve blood pressure regulation when standing. The first line of research utilized an analog of somatostatin to selectively constrict the splanchnic (gut) circulation, which dramatically increased the ability to maintain blood pressure while standing. Similarly, infusing a combination of fatty acids increased blood pressure by stimulating the sympathetic nervous system. The first study provdes sufficient information to warrant clinical trials in patients or a spaceflight analog environment. The second study is less developed, but suggests that dietary manipulation of fatty acid composition could improve blood pressure regulation when standing.
Task Progress:	In this project we focused on three main issues related to orthostatic tolerance: 1. Use of somatostatin analogs to improve orthostatic tolerance 2. Altering blood lipid concentrations to improve orthostatic tolerance 3. Developing a novel framework to characterize the hydrostatic indifferent point as a predictor or orthostatic tolerance Both clinical and empirical research has demonstrated that orthostatic tolerance (OT), the ability to maintain blood pressure in the standing position, is less in women than in men. This proposal related sex differences in OT to hydraulic properties of the circulation, which provides a theoretical and experimental framework to test a physiological and a pharmacological countermeasure. In a gravitational field, cardiac filling pressure is determined by the hydrostatic difference between the right atrium and the hydrostatic indifferent point (HIP), the point in the circulation where pressure does not change. The HIP is the hydraulic equivalent of the center of mass of a solid object. We hypothesized that the distribution of vascular volume in women results in an HIP more caudal than men, causing greater falls in filling pressure when assuming an upright position in a gravitational field. The location of the HIP is dependent on regional vascular compliance; in humans, characteristics of the highly compliant splanchnic and cutaneous circulations should therefore affect HIP and OT. These manipulations produce measurable cephalad shifts in HIP, raising cardiac filling pressure and increasing OT. A feature of the splanchnic circulation arguably more important to post-flight orthostatic intolerance is the high compliance of the splanchnic bed, which allows the splanchnic circulation to serve as a reservoir of blood through a combination of active venoconstriction and passive elastic recoil secondary to arterial vasoconstriction. Thus, the splanchnic circulation is a logical candidate to "autotransfuse" a hypovolemic and/or orthostatically intolerant subject with blood. Our proposal exploited this concept with a new class of pharmacological agents that have proven effective to treat orthostatic intolerance in some forms of autonomic dysfunction. Somatostatin analogs cause pronounced splanchnic with improvement in blood pressure regulation. We are pleased to report that the somatostatin intervention was largely successful. We used volunteers instrumented to record heart rate (ECG), beat-to-beat blood pressure (arterial tonometry), cardiac output (C2H2 rebreathing), splanchnic blood flow (ICG clearance), and limb blood flow (venous occlusion plethysmography) during 70° head-up tilt for 45 minutes or until presyncope. Based on clinical norms, this duration of tilt was designed to elicit a presyncopal response in 95% of women. Subjects repeated the test with and without 100µg octreotide infused over a 15-minute period. The study included a balanced, randomized, double blind, crossover design to control for potential ordering and placebo effects. This intervention proved to be remarkably efficacious. While no woman could tolerate 45 minutes of tilting in the control condition, half were able to do so following administration of the somatostatin analog. Octreotide administration produced dramatic increases in median tilt time. The fact that increases in tilt time were positively related to the reduction in splanchnic blood flow suggests that splanchnic vasoconstriction was largely responsible for maintaining cardiac output and vascular resistance. The data provided in this project relate to Bioastronautics Roadmap questions that address issues and countermeasures associated with diminished cardiac and vascular function countermeasures that are found in the Human Health and Countermeasures, Cardiovascular discipline of the Roadmap. Capitalizing on our earlier observations that reduction in total caloric intake would reduce orthostatic tolerance and sympathetic activity, we undertook a series of studies to manipulate sympathetic activity and orthostatic tolerance with acute increases in blood lipids using Intralipid® administration. Despite the emerging literature linking non-esterified fatty acids (NEFAs) to metabolic abnormalities and hypertension, there is a paucity of results regarding the role of the SNS (sympathetic nervous system) in the pressor response to NEFAs. This is surprising given the evidence that some forms of hypertension with obesity and high-fat diets may be

	neurogenically mediated. Therefore the main aim of this work was to investigate the role of NEFAs on central sympathetic activation in humans by direct measurement of MSNA (muscle sympathetic nerve activity) for a sufficient duration to establish a complete temporal response. On 2 days separated by at least 2 weeks, 17 lean healthy volunteers (ten male/seven female; age, 22 ±1 years; body mass index, 23±1 kg/m2) received a 4-h intravenous infusion of 20% Intralipid® or placebo (in a single-blind randomized balanced order). MSNA (muscle sympathetic nerve activity), HR (heart rate), blood pressure, and cardiac output were measured before and throughout infusion. The study demonstrated a remarkable pressor response to NEFA that was associated with an increase in SVR and MSNA. The data are suggestive that dietary manipulation of macronutrients might be another approach to improve orthostatic tolerance. To summarize, this study developed two different countermeasures (one pharmacologic, the other dietary) that could be used to improve blood pressure when standing.
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